# Introduction to Web APIs

ASP.NET Core Web API — Week 1 lecture notes

## 1. What is a Web API?

A Web API is a program that other programs talk to over HTTP. A React app, a mobile app or another server sends a **request** and the API sends back a **response**, usually as JSON. The API never draws a page; it only exchanges data.

## 2. HTTP in one page

Every request has a **method**, a **URL**, optional **headers** and an optional **body**. Every response has a **status code**, headers and a body.

| Method | Meaning | Example |
|---|---|---|
| GET | Read data | GET /api/courses |
| POST | Create something new | POST /api/courses |
| PUT | Replace / update | PUT /api/courses/5 |
| DELETE | Remove | DELETE /api/courses/5 |

## 3. REST: resources and URLs

• A **resource** is a noun: courses, students, payments.

• URLs name resources, methods say what to do: */api/courses/5* + DELETE.

• Use plural nouns and ids: /api/courses, /api/courses/{id}.

• The server keeps no conversation state; each request carries what it needs (for example a JWT token).

## 4. A typical JSON exchange

```
GET /api/courses/2

HTTP/1.1 200 OK
Content-Type: application/json

{
  "id": 2,
  "title": "ASP.NET Core Web API",
  "price": 1500,
  "status": "OpenForEnrollment"
}
```

## 5. How an ASP.NET Core project is organised

| Folder / file | Responsibility |
|---|---|
| Program.cs | Builds the app: services (DI), middleware, routing |
| Controllers/ | Receive HTTP requests, return responses |
| Services/ | Business rules (validation, guards) |

| | |
|---|---|
| Repos/ | Talk to the database through EF Core |
| Models/ | Classes that become database tables |
| DTOs/ | Shapes of data sent and received by the API |

A request travels **Controller** → **Service** → **Repository** → **Database** and the answer travels back. Keeping these layers separate makes the code easy to test and change.

## 6. Your first endpoint

```
[ApiController]
[Route("api/[controller]")]
public class HelloController : ControllerBase
{
    [HttpGet]
    public IActionResult Get() => Ok(new { message = "Hello from the API" });
}
```

## Checklist for this week

- Create a new Web API project and run it.
- Open Swagger and call your endpoint.
- Read the official tutorial linked in the course materials.